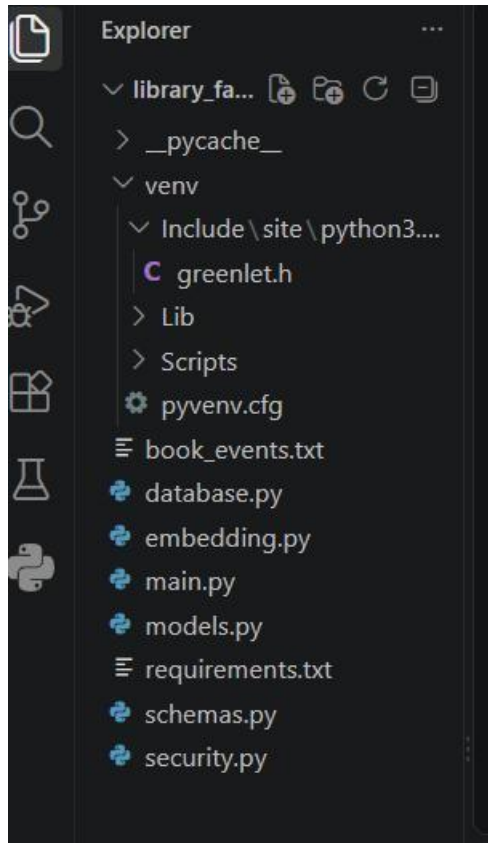


Practical-5

2520090176b-Y Mohana Priya

1. Project Structure



2. Installed Packages

```

File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\starlette\routing.py", line 648, in lifespan
    async with self.lifespan_context(app) as maybe_state:
File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\fastapi\routing.py", line 265, in __aenter__
    await self.router.startup()
File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\fastapi\routing.py", line 6375, in _startup
    handler()
File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\fastapi\main.py", line 38, in startup
    async with engine.begin() as connection:
File "C:\Users\satwika\AppData\Local\Programs\Python\Python311\Lib\contextlib.py", line 210, in __aenter__
    return await anext(self.gen)
    ^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\sqlalchemy\ext\asyncio\engine.py", line 1068, in
    begin with conn:
File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\sqlalchemy\ext\asyncio\base.py", line 121, in
    __aenter__,
    return await self.start(is_ctxmanager=True)
File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\sqlalchemy\engine\base.py", line 144, in __init__
    self._dbapi_connection = engine.raw_connection()
    ^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\sqlalchemy\engine\base.py", line 3319, in raw_connection
    return self.pool.connect()
    ^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\sqlalchemy\pool\base.py", line 448, in connect
    return _ConnectionFairy._checkout(self)
    ^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\sqlalchemy\pool\base.py", line 1272, in _checkout
    fairy = _ConnectionRecord.checkout(pool)
    ^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\sqlalchemy\pool\base.py", line 712, in checkout
    rec = pool._do_get()
    ^^^^^^^^^^^^^^^^^^^^^

```

```
File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\starlette\routing.py", line 648, in lifespan
    async with self.lifespan_context(app) as maybe_state:
    ,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,

File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\fastapi\routing.py", line 265, in __aenter__
    await self.router.startup()
    ,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,

File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\fastapi\main.py", line 38, in startup
    async with engine.begin() as connection:
    ,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,

File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\sqlalchemy\pool\base.py", line 1272, in _enter_
    return await _create_connection()
    ,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,

File "C:\Users\satwika\OneDrive\Desktop\DBBS\Library_fastapi_project\venv\Lib\site-packages\sqlalchemy\pool\base.py", line 712, in _checkout
    rec = pool._do_get()
    ,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,
```

The screenshot displays the MySQL Workbench interface. The top toolbar includes icons for file operations, database management, and execution. The left sidebar contains the 'MANAGEMENT' and 'INSTANCE' sections. The central SQL Editor shows the creation of a database named 'library_db' and a table named 'members' with columns: 'id' (INT AUTO_INCREMENT PRIMARY KEY), 'username' (VARCHAR(100) UNIQUE NOT NULL), 'password_hash' (VARCHAR(255) NOT NULL), and 'role' (VARCHAR(30) DEFAULT 'member'). The bottom section shows the 'Result Grid' with a list of databases, including 'bookflow_db', 'bookflow_dh', 'information_schema', 'kludb', 'library_db', 'madhulika_2520090056', 'mysql', and 'performance_schema'. The 'Output' tab at the bottom shows the execution results of the SQL statements, including the creation of the database and table, and the execution of the 'SHOW DATABASES;' command.

The screenshot shows a database management tool interface with a sidebar on the left containing sections for MANAGEMENT, INSTANCE, and PERFORMANCE. The main window displays SQL code for creating a table and executing queries. Below the code, the 'Result Grid' shows the tables in the 'library_db' database: 'books' and 'members'. The 'Output' pane at the bottom shows the execution log for these queries.

SQL Code:

```

10 id INT AUTO_INCREMENT PRIMARY KEY,
11 title VARCHAR(255) NOT NULL,
12 author VARCHAR(255) NOT NULL,
13 isbn VARCHAR(13) UNIQUE NOT NULL,
14 price DECIMAL(10,2) NOT NULL,
15 published_date DATE NOT NULL,
16 description TEXT,
17 embedding JSON
18 );
19 SHOW DATABASES;
20 USE library_db;
21 SHOW TABLES;

```

Result Grid:

Tables_in_library_db
books
members

Output:

#	Time	Action	Message
4	20:05:32	CREATE TABLE books (id INT AUTO_INCREMENT PRIM...	0 row(s) affected
5	20:05:55	SHOW DATABASES	10 row(s) returned
6	20:07:46	USE library_db	0 row(s) affected
7	20:07:52	SHOW TABLES	2 row(s) returned

4. Database.py

The screenshot shows a code editor with the 'database.py' file open. The code defines the database URL, creates an async engine, sets up an async sessionmaker, and defines a base class for the database models.

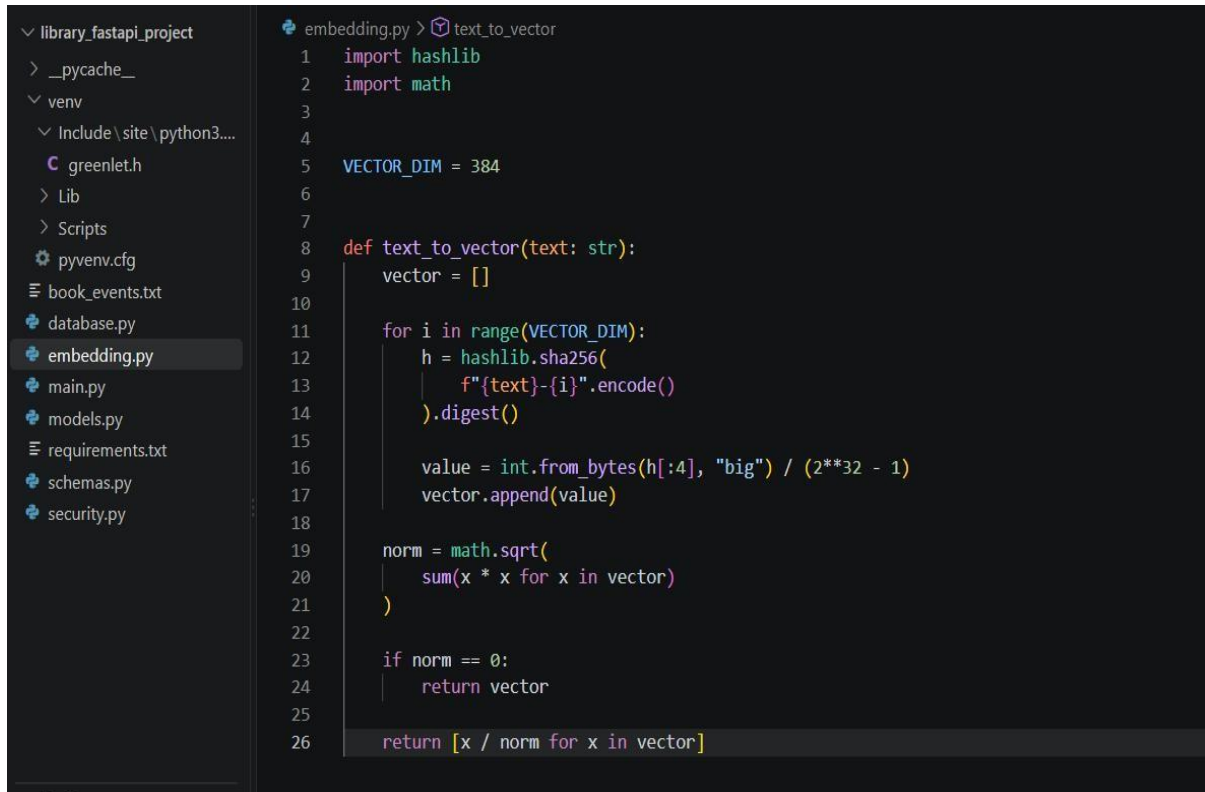
Code:

```

1 from sqlalchemy.ext.asyncio import AsyncSession, async_sessionmaker, create_async_engine
2 from sqlalchemy.orm import DeclarativeBase
3
4 DATABASE_URL = (
5     "mysql+asyncmy://root:Madhu%40248"
6     "@localhost:3306/library_db"
7 )
8
9 engine = create_async_engine(
10     DATABASE_URL,
11     echo=True
12 )
13
14 AsyncSessionLocal = async_sessionmaker(
15     engine,
16     expire_on_commit=False
17 )
18
19
20 class Base(DeclarativeBase):
21     pass
22
23
24 async def get_db():
25     async with AsyncSessionLocal() as session:
26         yield session

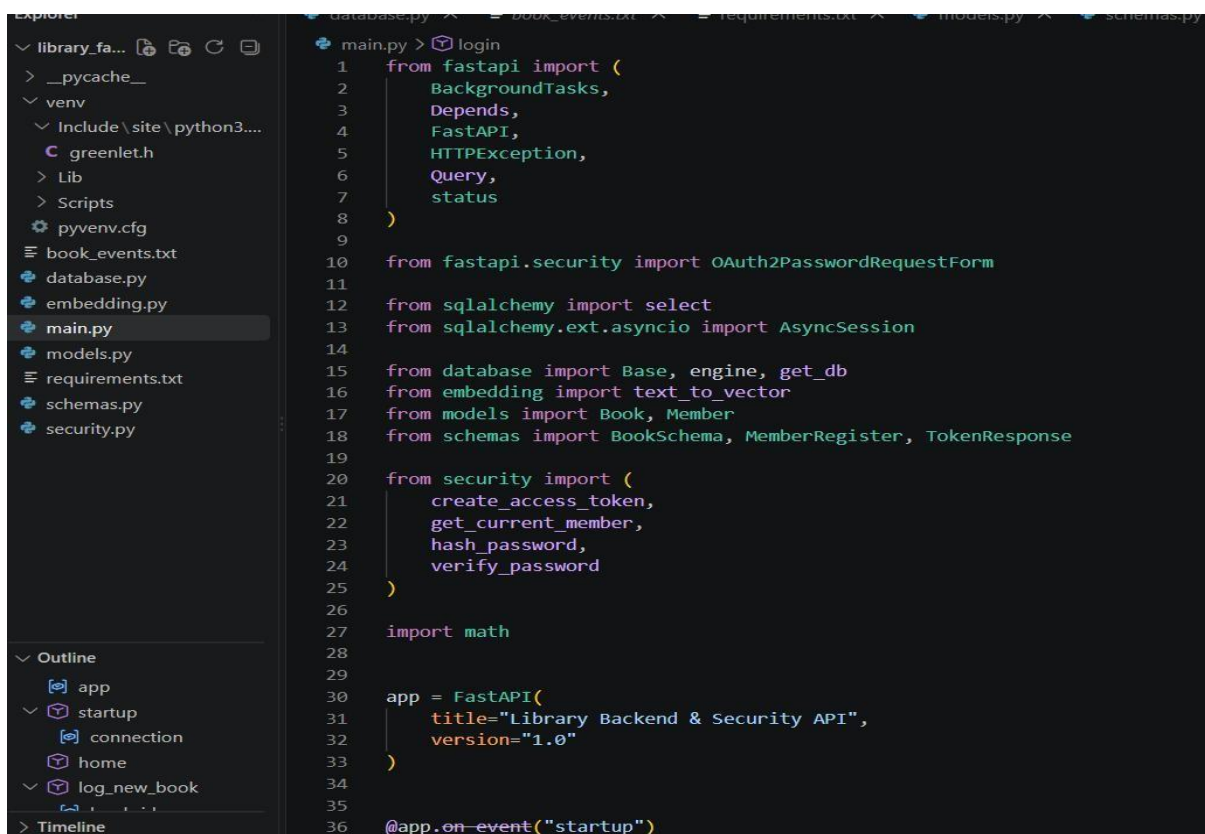
```

5. Embedding.py



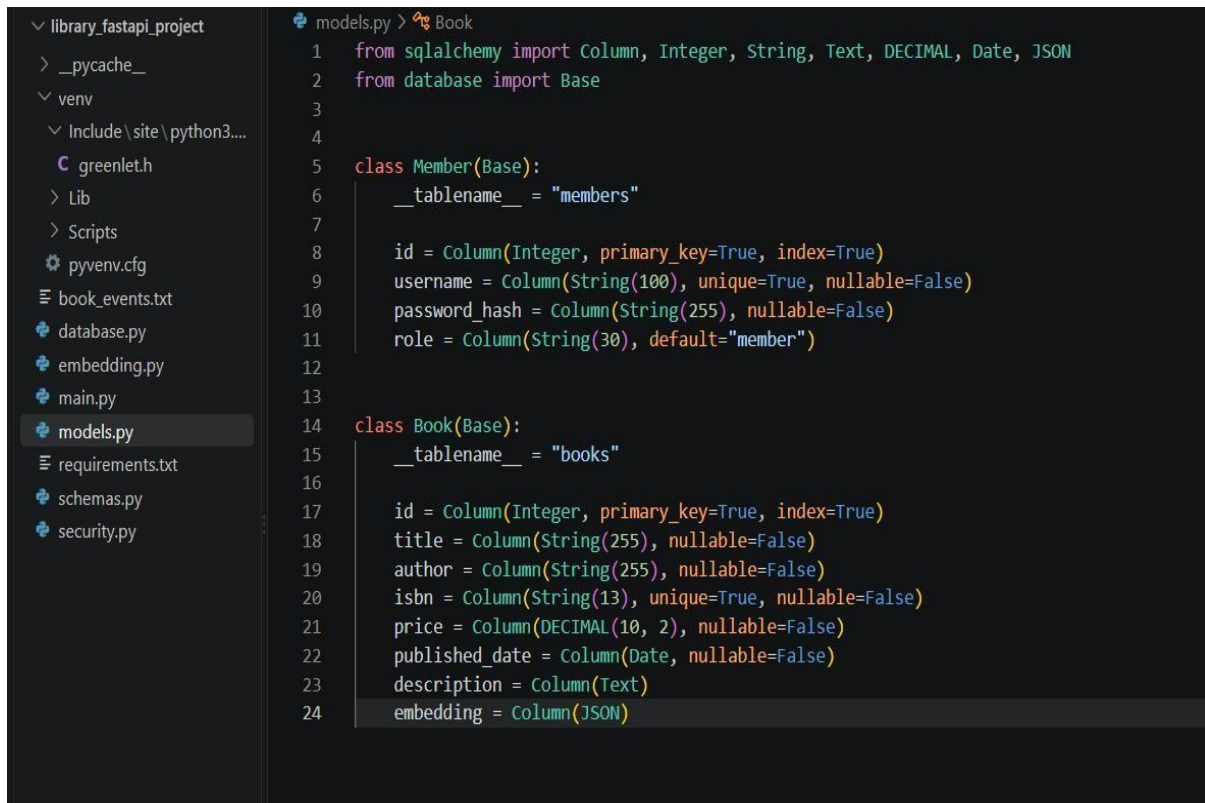
```
1 import hashlib
2 import math
3
4
5 VECTOR_DIM = 384
6
7
8 def text_to_vector(text: str):
9     vector = []
10
11     for i in range(VECTOR_DIM):
12         h = hashlib.sha256(
13             f"{text}-{i}".encode()
14         ).digest()
15
16         value = int.from_bytes(h[:4], "big") / (2**32 - 1)
17         vector.append(value)
18
19     norm = math.sqrt(
20         sum(x * x for x in vector)
21     )
22
23     if norm == 0:
24         return vector
25
26     return [x / norm for x in vector]
```

6. Main.py



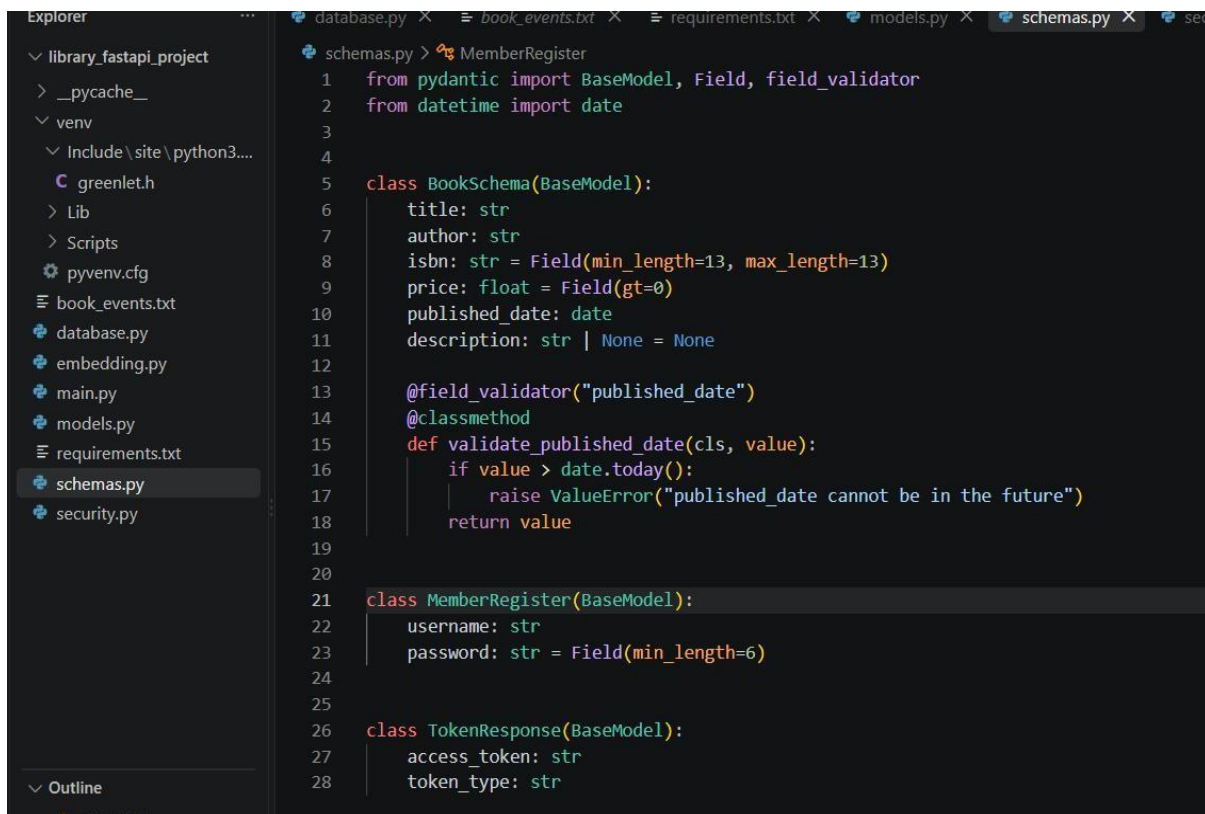
```
1 from fastapi import (
2     BackgroundTasks,
3     Depends,
4     FastAPI,
5     HTTPException,
6     Query,
7     status
8 )
9
10 from fastapi.security import OAuth2PasswordRequestForm
11
12 from sqlalchemy import select
13 from sqlalchemy.ext.asyncio import AsyncSession
14
15 from database import Base, engine, get_db
16 from embedding import text_to_vector
17 from models import Book, Member
18 from schemas import BookSchema, MemberRegister, TokenResponse
19
20 from security import (
21     create_access_token,
22     get_current_member,
23     hash_password,
24     verify_password
25 )
26
27 import math
28
29
30 app = FastAPI(
31     title="Library Backend & Security API",
32     version="1.0"
33 )
34
35
36 @app.on_event("startup")
```

7. Models.py



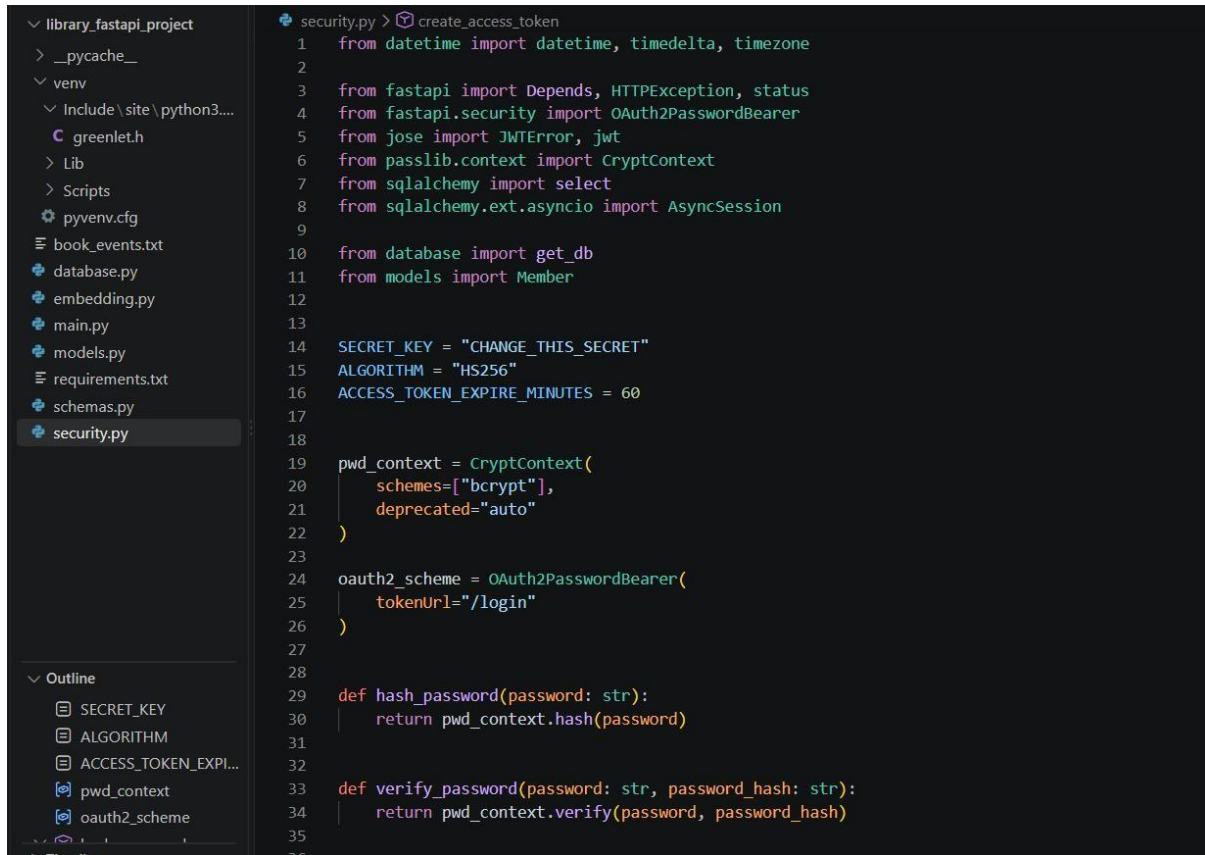
```
models.py > Book
1  from sqlalchemy import Column, Integer, String, Text, DECIMAL, Date, JSON
2  from database import Base
3
4
5  class Member(Base):
6      __tablename__ = "members"
7
8      id = Column(Integer, primary_key=True, index=True)
9      username = Column(String(100), unique=True, nullable=False)
10     password_hash = Column(String(255), nullable=False)
11     role = Column(String(30), default="member")
12
13
14 class Book(Base):
15     __tablename__ = "books"
16
17     id = Column(Integer, primary_key=True, index=True)
18     title = Column(String(255), nullable=False)
19     author = Column(String(255), nullable=False)
20     isbn = Column(String(13), unique=True, nullable=False)
21     price = Column(DECIMAL(10, 2), nullable=False)
22     published_date = Column(Date, nullable=False)
23     description = Column(Text)
24     embedding = Column(JSON)
```

8. Schemas.py



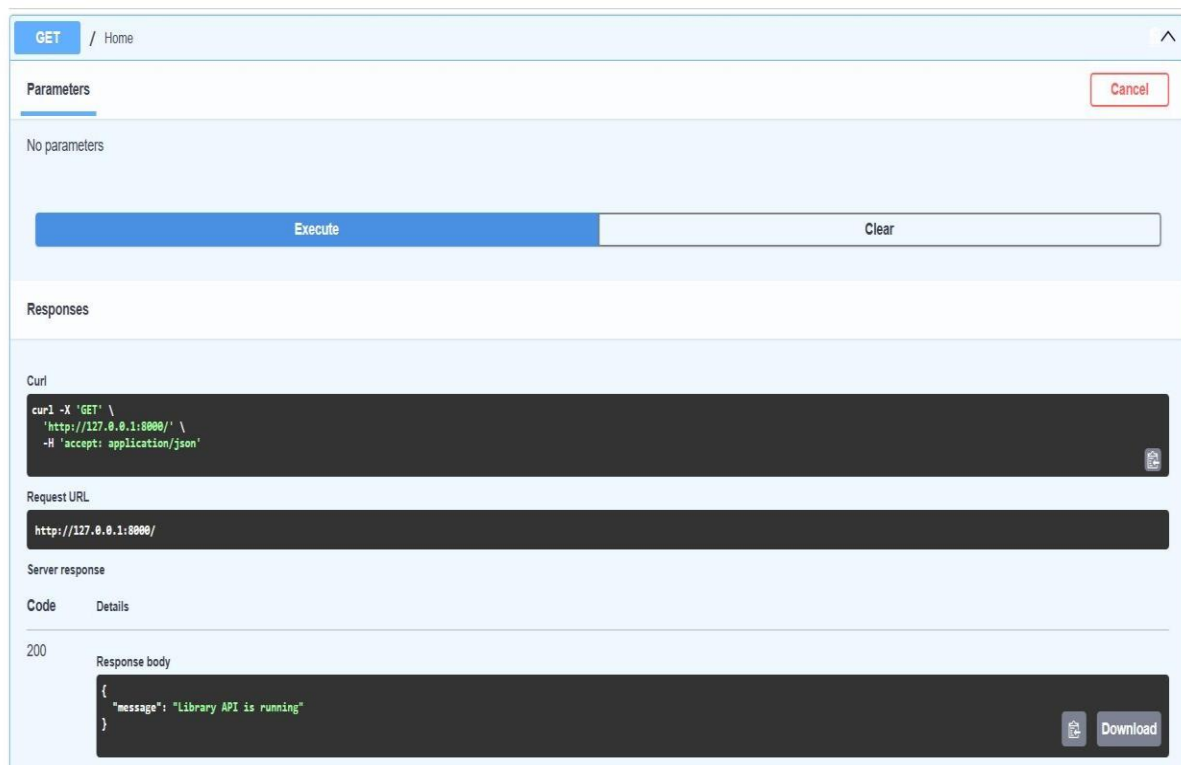
```
schemas.py > MemberRegister
1  from pydantic import BaseModel, Field, field_validator
2  from datetime import date
3
4
5  class BookSchema(BaseModel):
6      title: str
7      author: str
8      isbn: str = Field(min_length=13, max_length=13)
9      price: float = Field(gt=0)
10     published_date: date
11     description: str | None = None
12
13     @field_validator("published_date")
14     @classmethod
15     def validate_published_date(cls, value):
16         if value > date.today():
17             raise ValueError("published_date cannot be in the future")
18         return value
19
20
21 class MemberRegister(BaseModel):
22     username: str
23     password: str = Field(min_length=6)
24
25
26 class TokenResponse(BaseModel):
27     access_token: str
28     token_type: str
```

9. Security.py



```
security.py > create_access_token
1  from datetime import datetime, timedelta, timezone
2
3  from fastapi import Depends, HTTPException, status
4  from fastapi.security import OAuth2PasswordBearer
5  from jose import JWTError, jwt
6  from passlib.context import CryptContext
7  from sqlalchemy import select
8  from sqlalchemy.ext.asyncio import AsyncSession
9
10 from database import get_db
11 from models import Member
12
13
14 SECRET_KEY = "CHANGE_THIS_SECRET"
15 ALGORITHM = "HS256"
16 ACCESS_TOKEN_EXPIRE_MINUTES = 60
17
18
19 pwd_context = CryptContext(
20     schemes=["bcrypt"],
21     deprecated="auto"
22 )
23
24 oauth2_scheme = OAuth2PasswordBearer(
25     tokenUrl="/login"
26 )
27
28
29 def hash_password(password: str):
30     return pwd_context.hash(password)
31
32
33 def verify_password(password: str, password_hash: str):
34     return pwd_context.verify(password, password_hash)
35
36
```

10. Test Registration



POST/registerRegister

Parameters

CancelReset

No parameters

Request bodyrequired

application/json

Edit ValueSchema

```
{
  "username": "Madhu",
  "password": "library123"
}
```

Execute

Clear

Responses

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/register' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "username": "Madhu",
    "password": "library123"
  }'
```

Request URL

http://127.0.0.1:8000/register

Server response

CodeDetails

200

Response body

```
{
  "message": "Member registered successfully",
  "username": "Madhu",
  "role": "member"
}
```

Download

Response headers

```
content-length: 79
content-type: application/json
date: Mon, 14 Sep 2026 16:05:16 GMT
server: uvicorn
```

Responses

11. Test Login

POST /login Login

Parameters

Cancel

Reset

No parameters

Request body required

application/x-www-form-urlencoded

grant_type

string | (string | null)

pattern: ^password\$

☐ Send empty value

username required

string

password required

string(\$password)

scope

string

☒ Send empty value

client_id

string | (string | null)

☐ Send empty value

client_secret

string | (string | null)(\$password)

☐ Send empty value

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/login' \
  -H 'accept: application/json' \
  -H 'Content-type: application/x-www-form-urlencoded' \
  -d 'grant_type=password&username=Madhu&password=library123&scope=&client_id=Madhu&client_secret=library123'
```

Request URL

http://127.0.0.1:8000/login

Server response

Code	Details
200	Response body { "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IWRXVjQ9LmVzdmI0IjNYMmR0dSIsInR5cGU0Ij0iZmVlZC1lLCJleHAiOjE3ODk0MDU5Nzh9,BSyLTpIyOmAoCo0hwV3PUg1xVb9I2uZ8d08fy3Vx4", "token_type": "bearer" } Download Response headers content-length: 186 content-type: application/json date: Mon, 14 Sep 2026 16:12:57 GMT server: uvicorn

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header

Example Value | Schema

12. Test POST/Books

POST

/books Create Book

🔒 🔗

Parameters

Cancel

Reset

No parameters

Request body required

application/json

Edit Value

Schema

```
{
  "title": "Python for Beginners",
  "author": "John Smith",
  "isbn": "123",
  "price": -100.00
  "published_date": "2024-06-10",
  "description": "An introduction to Python programming."
}
```

Execute

Clear

Responses

```
-H 'Host: 127.0.0.1:8000/books' \
-H 'accept: application/json' \
-H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJJYXRodSIsInVubGUoiOiJtZWUiLCJleHAiOjE3ODk0MDYyMjZ9.Lq7z0UsTHMKWxQHBzXEgTX64tTGK1sHhu4JgWnB1l0' \
-H 'Content-Type: application/json' \
-d '{
    "title": "Python for Beginners",
    "author": "John Smith",
    "isbn": "123",
    "price": 450.00,
    "published_date": "2024-06-10",
    "description": "An introduction to Python programming."
}'
```

Request URL

```
http://127.0.0.1:8000/books
```

Server response

Code	Details
422	Error: Unprocessable Entity

Response body

```
{
  "detail": [
    {
      "type": "string_too_short",
      "loc": [
        "body",
        "isbn"
      ],
      "msg": "String should have at least 13 characters",
      "input": "123",
      "ctx": {
        "min_length": 13
      }
    }
  ]
}
```

Response headers

```
content-length: 150
content-type: application/json
date: Mon, 14 Sep 2026 16:28:41 GMT
server: uvicorn
```

GET

/books/search

Search Books

Parameters

Cancel

Name	Description
q * required	
string	q = python
(query)	minLength: 1
limit	
integer	5
(query)	maximum: 20 minimum: 1

ExecuteClear

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/books/search?q=q120K30K120python&limit=5' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/books/search?q=q120K30K120python&limit=5
```

Server response

13. book_event.txt

library_fastapi_project

> __pycache__

venv

Include \site\python3....

greenlet.h

> Lib

> Scripts

pyvenv.cfg

book_events.txt

database.py

embedding.py

main.py

models.py

requirements.txt

schemas.py

security.py

book_events.txt

1 New Book Added: id=1, title=Python for Beginners

2

